MATERIAL SCIENCE · SEMESTER 1 · UNIT 3

Unit 3 Review: Metals

Name _____

Date _____ Period _____

Show your work for calculations and use full sentences for explanations. Answers are in the answer key that comes with this worksheet in Classboard.

A · BONDING AND STRUCTURE

1. Explain why copper wire can be bent into a loop but a ceramic mug shatters when dropped. Use the word "bond" in your answer.

2. Match each metal to its usual crystal structure (BCC, FCC or HCP): iron at room temperature, aluminum, zinc.

3. Sample X has large grains and sample Y has tiny grains. Same metal. Which is stronger, and why?

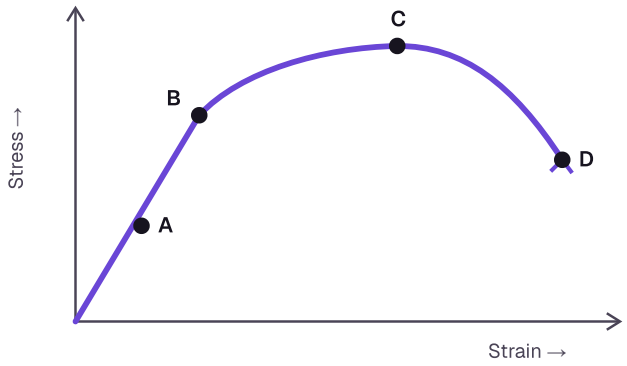
B · STRESS AND STRAIN

4. A bar with a cross-section of 60 mm² carries 12,000 N. Find the stress in MPa.

5. The same bar was 150.0 mm long and stretched to 150.3 mm (still elastic). Find the strain.

6. Use your answers to 4 and 5 to find Young's modulus.

7. Label points A–D on the curve: yield point, fracture, ultimate tensile strength, a point in the elastic region.



C · HEAT TREATMENT AND WORK HARDENING

8. Three bobby pins are heated red-hot. Pin 1 cools slowly in air. Pin 2 is plunged into water. Pin 3 is plunged into water, then reheated gently and cooled. Predict what each does when you try to bend it.

9. Why does a paper clip break after you bend it back and forth many times?

D · ALLOYS AND MELTING

10. Name what each alloy is made of: brass, bronze, steel, sterling silver.

11. Is steel a substitutional or interstitial alloy? Explain.

12. Tin melts at 232 °C and lead at 327 °C. Why do electronics use 63% tin / 37% lead solder (melting point 183 °C) instead of either pure metal? What is this mix called?

E · MAKING AND PROTECTING METALS

13. Number these lost-wax casting steps in order: __ pour molten metal · __ carve the wax model · __ burn out the wax · __ break off the plaster · __ cover the model in plaster

14. Give two ways to keep a steel railing outside from rusting, and explain how one of them works.

F · CLAIM, EVIDENCE, REASONING

15. A blacksmith quenches a knife blade and then tempers it. Write a claim about why both steps are needed, give evidence (from the bobby pin lab or the stress-strain curve), and explain your reasoning.